NOTE: There is no part 1 and part 2 of the program, you can evaluate it accordingly.

Program Overview:

- The program calculates the average, mode, and median of a given array of numbers within a specified interval.

- It allows the user to add new numbers to the array and recalculates the average, mode, and median based on the updated array.

- The program provides statistical analysis of the numbers within a specified interval, including average, mode, and median.

- It allows for dynamic addition of new numbers to the dataset for real-time analysis.

Program Functions:

1. `calculate_average(int arr[], int n)`:

- Calculates the average of the numbers in the array.

- Parameters:

  - `arr[]`: The array of numbers.

  - `n`: The number of elements in the array.

2. `calculate_mode(int arr[], int n, int a, int b)`:

- Calculates the mode (most frequently occurring number) within a specified range `[a, b]`.

- Parameters:

  - `arr[]`: The array of numbers.

  - `n`: The number of elements in the array.

- `a`: Lower bound of the range.

- `b`: Upper bound of the range.

3. `calculate_median(int arr[], int n)`:

- Calculates the median of the numbers in the array.

- Uses a sorting algorithm to rearrange the array in ascending order.

- Parameters:

  - `arr[]`: The array of numbers.

  - `n`: The number of elements in the array.

Code Usage:

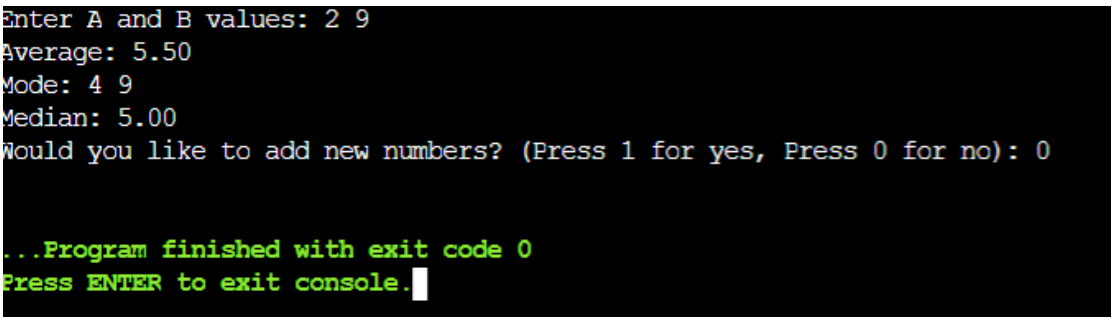
1. The user inputs the lower and upper bounds of the interval `[a, b]`.

2. The program calculates and displays the initial average, mode, and median.

3. The user can choose to add new numbers to the array and recalculate the statistics.

4. The program displays the updated average, mode, and median based on the modified array.

OUTPUTS:



```
Enter A and B values: 2 9
Average: 5.50
Mode: 4 9
Median: 5.00
Would you like to add new numbers? (Press 1 for yes, Press 0 for no): 1
Enter a new number: 5 6 -1
Average: 5.47
Mode: 4 5 9
Median: 5.00
```